

Name: M.SAIM MUSTAFA

ROLL#: 23-NTU-CS-1271

Assignment No: 1 Part B

Course Name: Embedded IoT

Semester: 5th

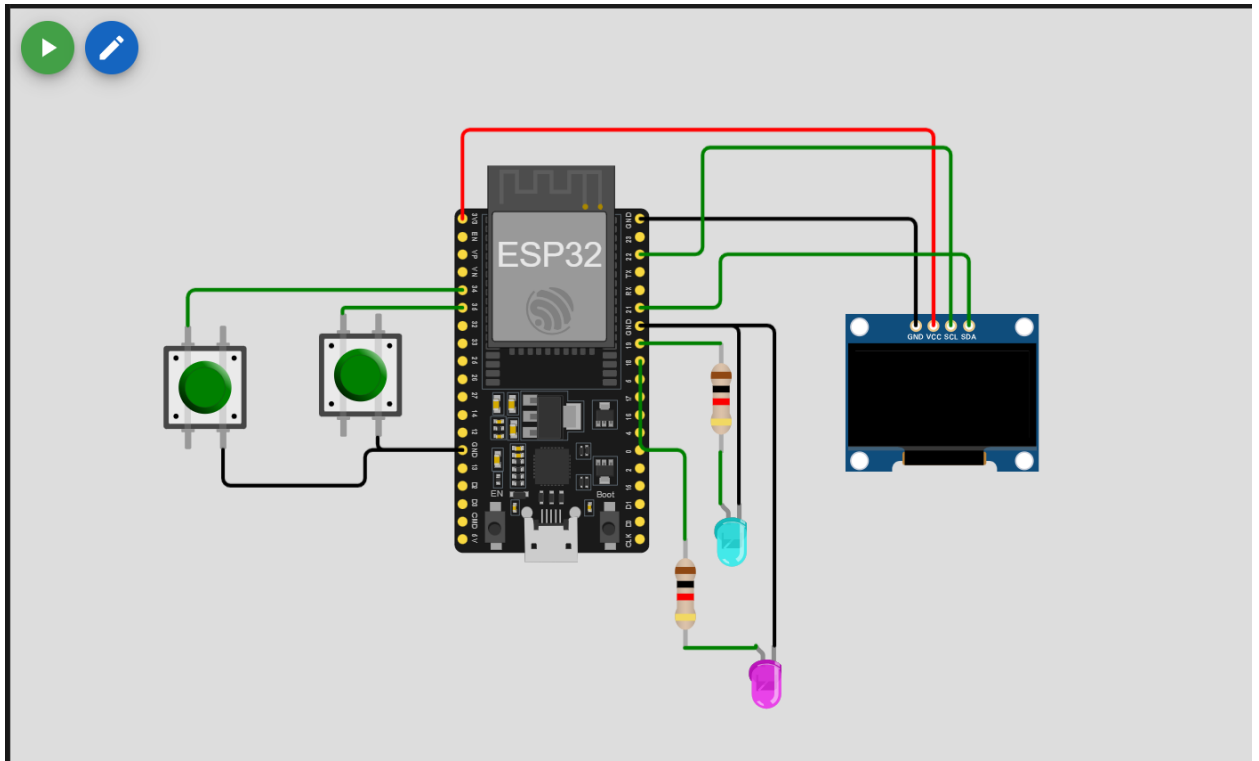
Section: BSAI

Date: 26 OCT,2025

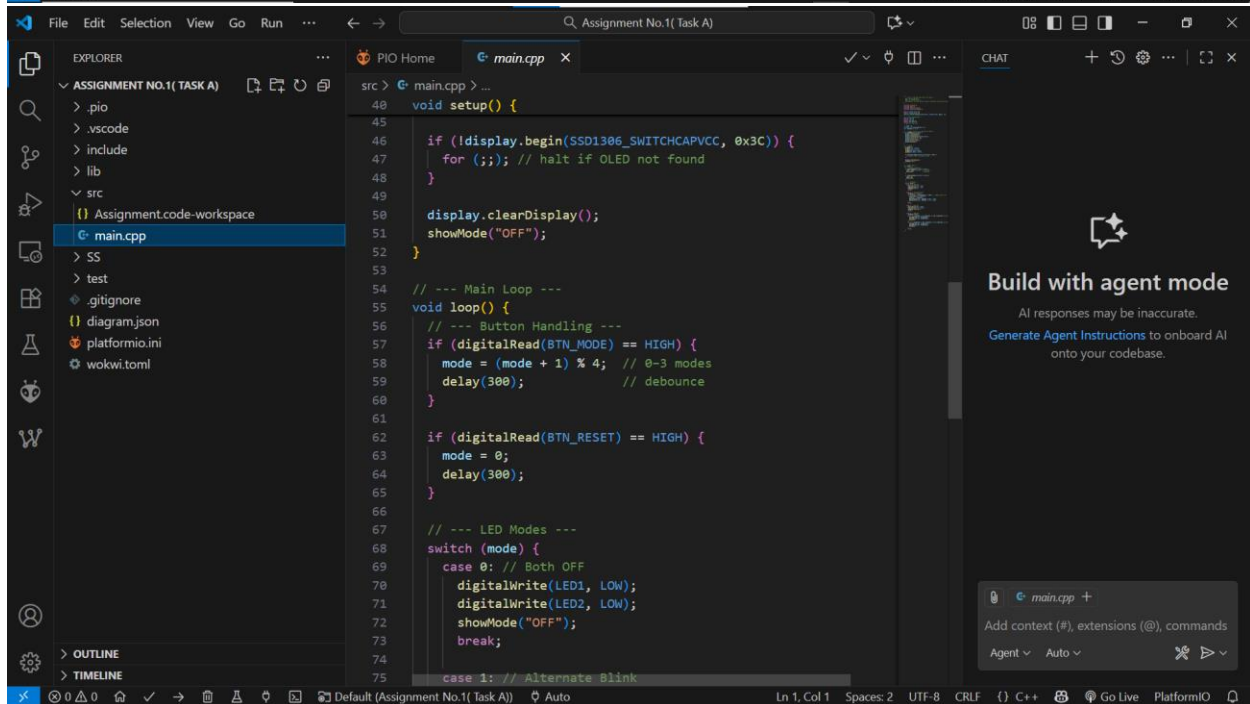
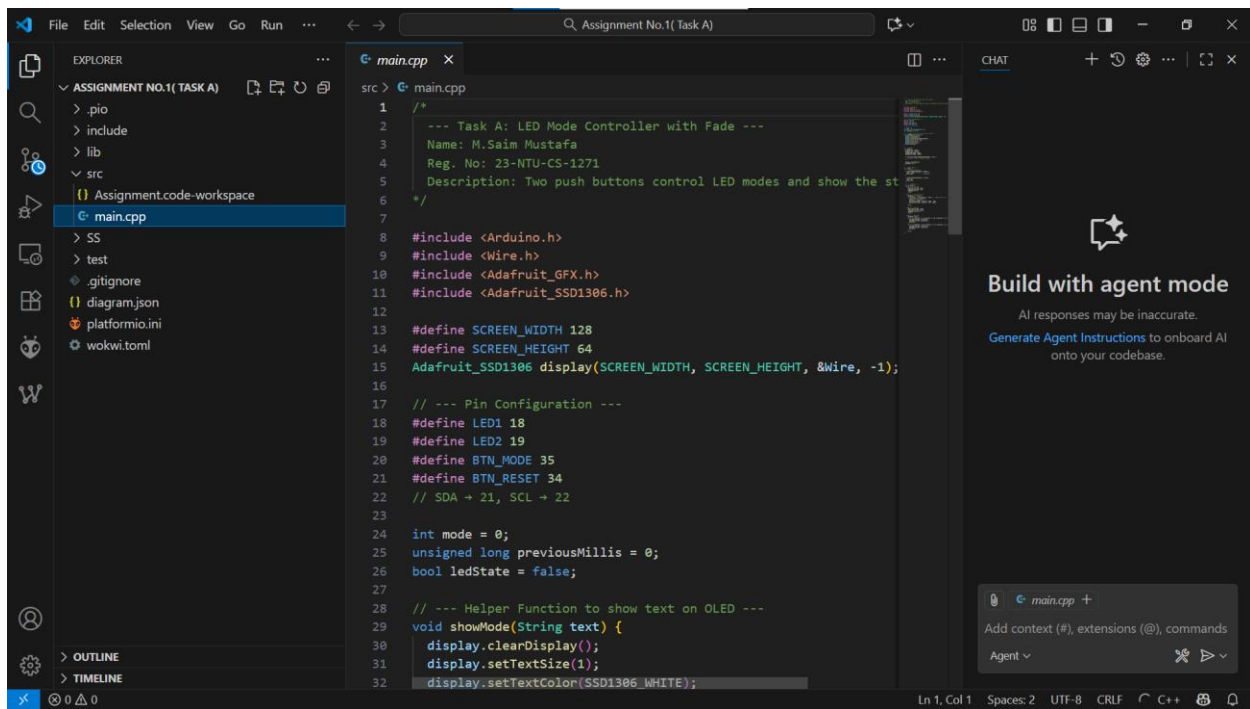
Question 3

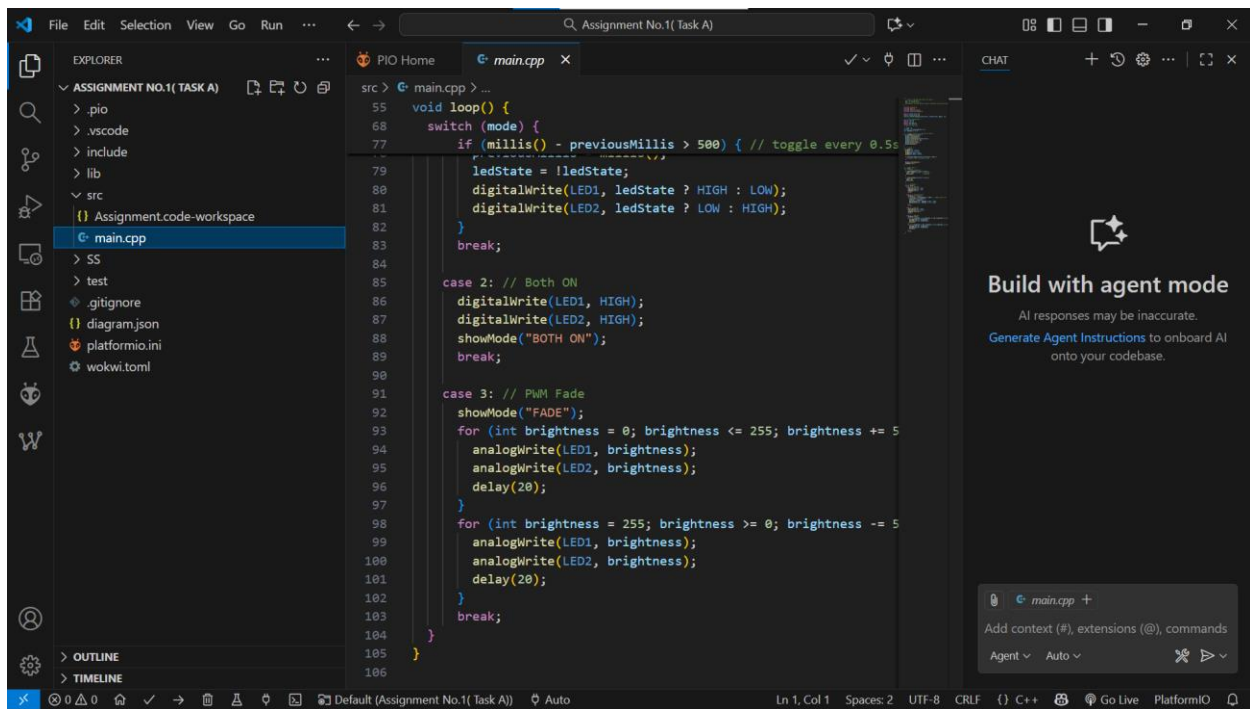
TASK A:

➤ Circuit Diagrams:



➤ Code Snippets:

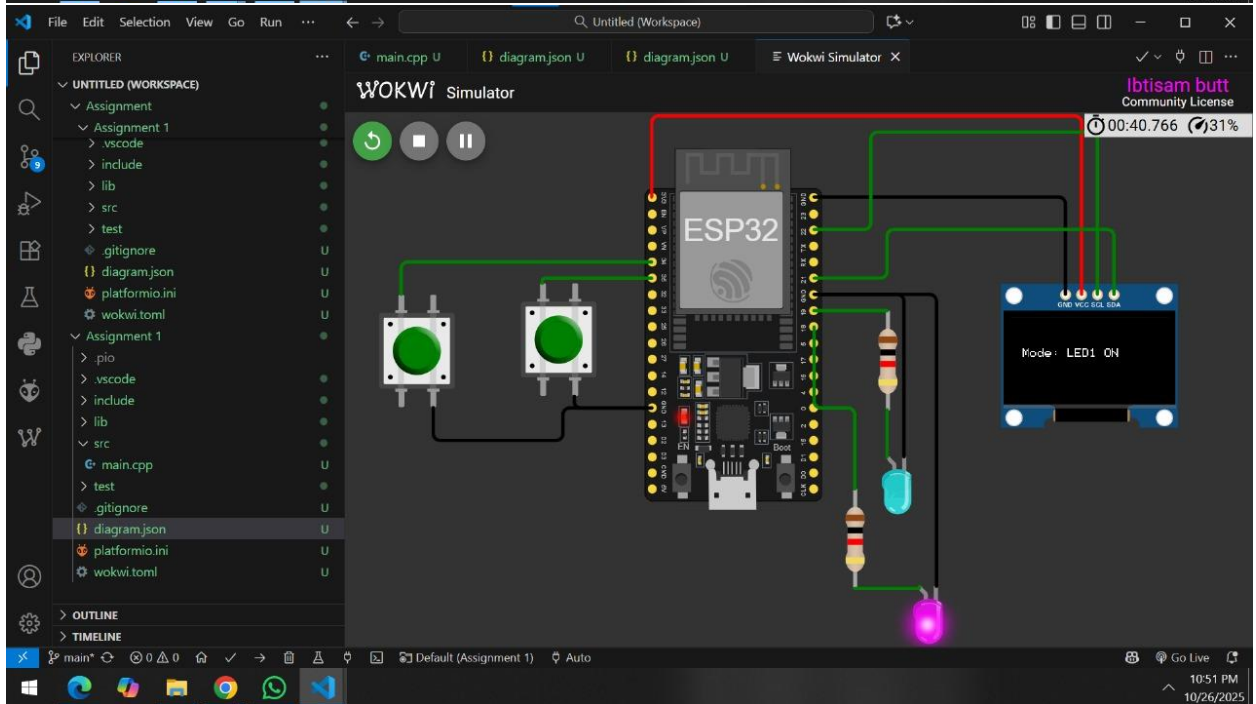
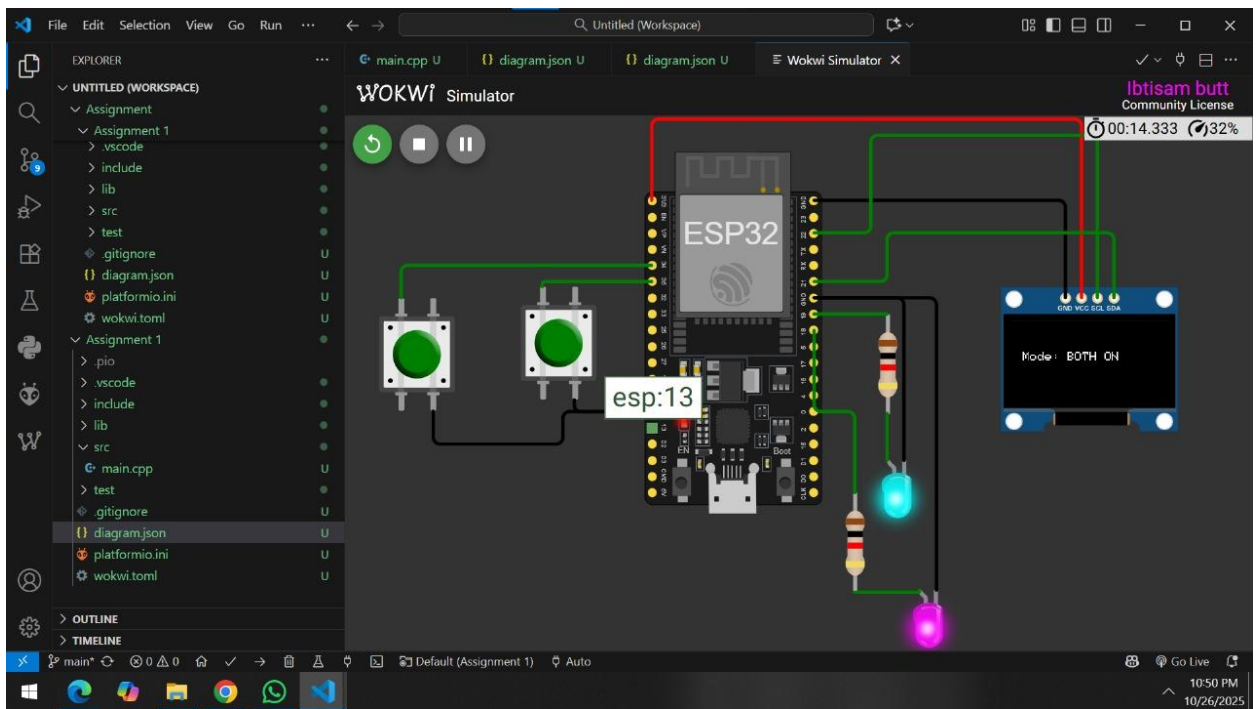


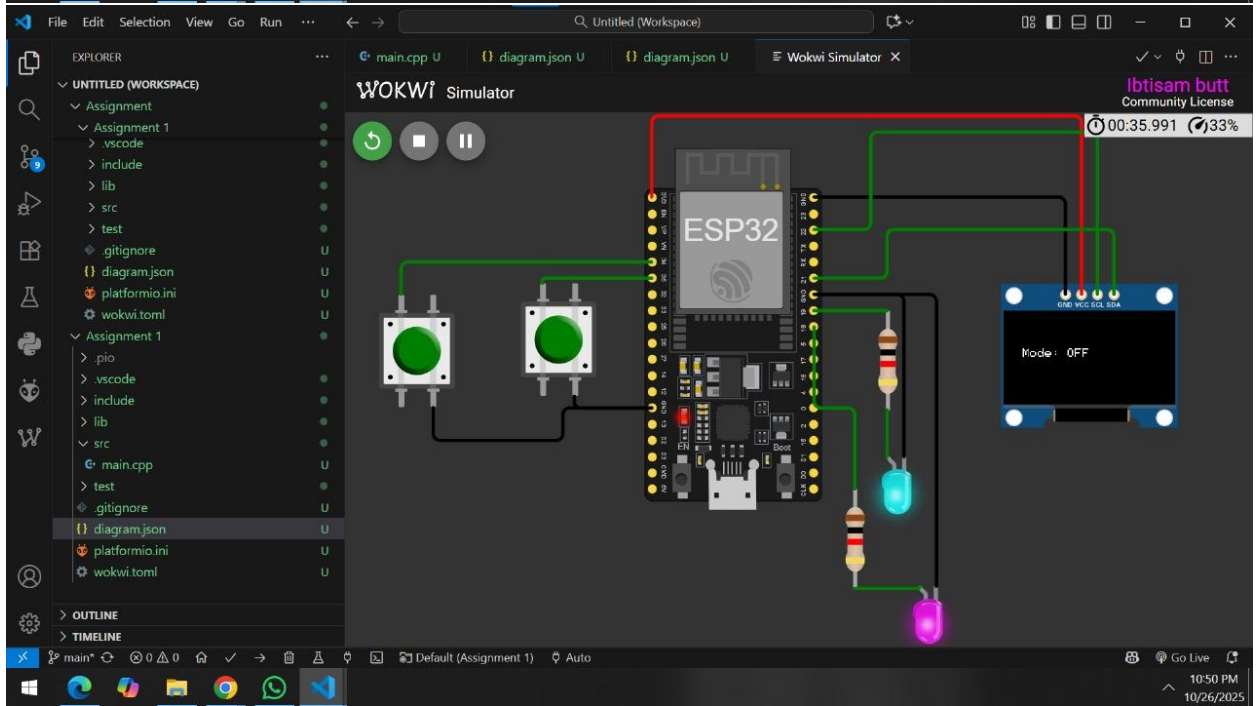
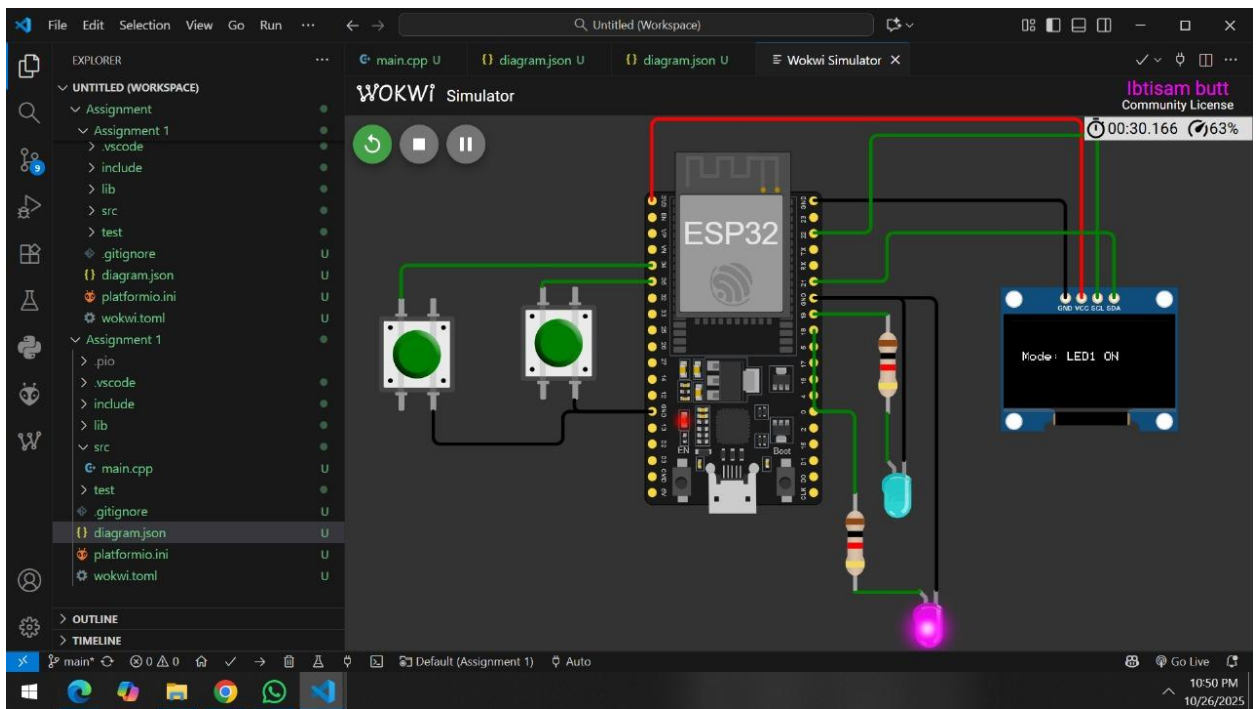


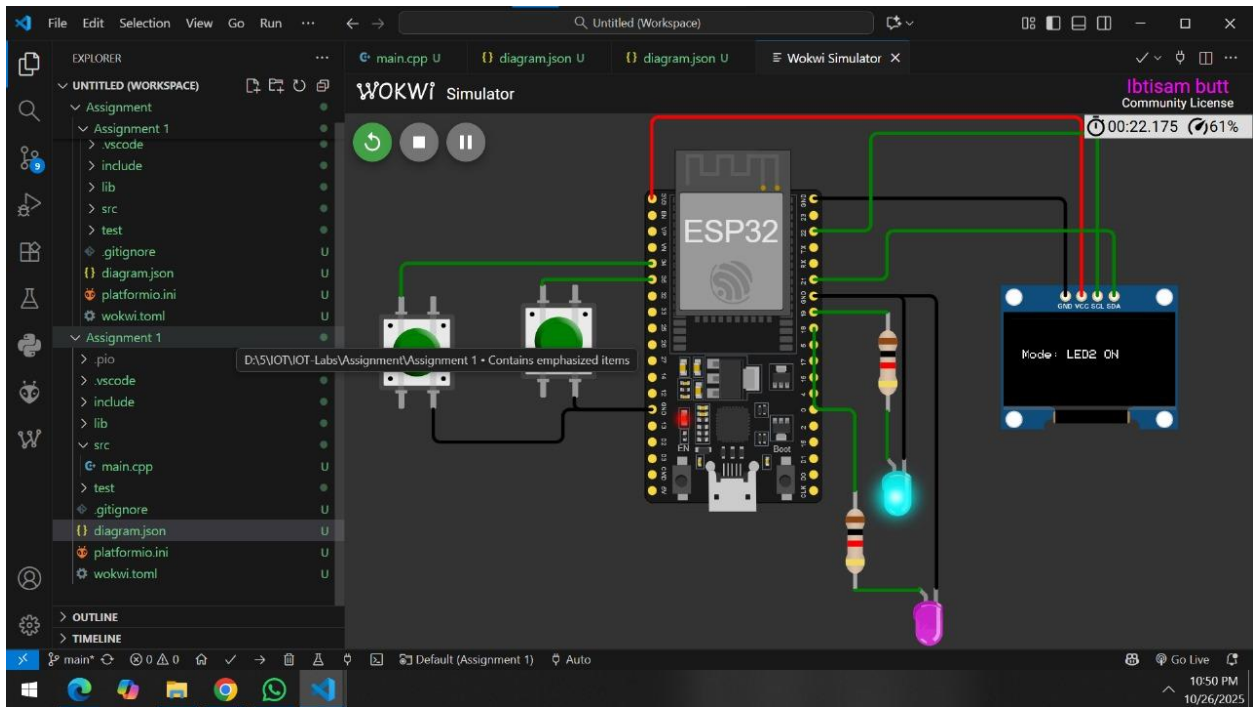
➤ Pin Map:

```
## Pin Map
- LED1 → GPIO 18
- LED2 → GPIO 19
- Button1 → GPIO 35
- Button2 → GPIO 34
- OLED → SDA 21, SCL 22
```

➤ Screenshots:

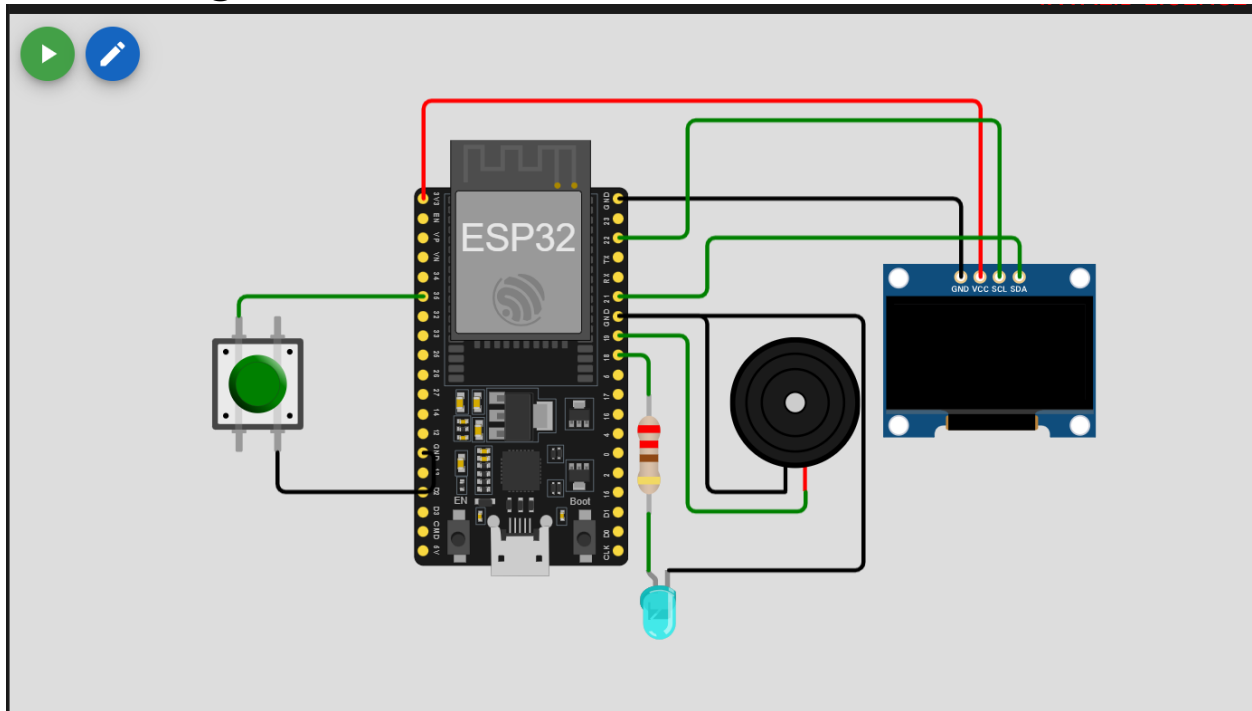






TASK B:

Circuit Diagrams:



➤ Code Snippets:

```
File Edit Selection View Go Run ... Assignment No.1 (Task B)

EXPLORER
  ASSIGNMENT NO.1 (TASK B)
    > .pio
    > include
    > lib
    > src
      Assignment.code-workspace
      main.cpp
    > SS
    > test
    > .gitignore
    > diagram.json
    > platformio.ini
    > wokwi.toml

main.cpp
1  /*
2  --- Task B: Button Press Detection ---
3  Name: M.Saim Mustafa
4  Reg. No: 23-NTU-CS-1271
5  Description: Short press toggles LED, long press triggers buzzer
6  */
7
8  #include <Arduino.h>
9  #include <Wire.h>
10 #include <Adafruit_GFX.h>
11 #include <Adafruit_SSD1306.h>
12
13 #define SCREEN_WIDTH 128
14 #define SCREEN_HEIGHT 64
15 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
16
17 // --- Pin Configuration ---
18 #define LED_PIN 18
19 #define BUTTON_PIN 35
20 #define BUZZER_PIN 19 // connect buzzer here
21
22 unsigned long buttonPressTime = 0;
23 bool buttonHeld = false;
24 bool ledState = false;
25
26 // --- Helper Function to show text on OLED ---
27 void showEvent(String text) {
28   display.clearDisplay();
29   display.setTextSize(1);
30   display.setTextColor(SSD1306_WHITE);
31   display.setCursor(10, 25);
32   display.println(text);
33 }
```

Build with agent mode

AI responses may be inaccurate.

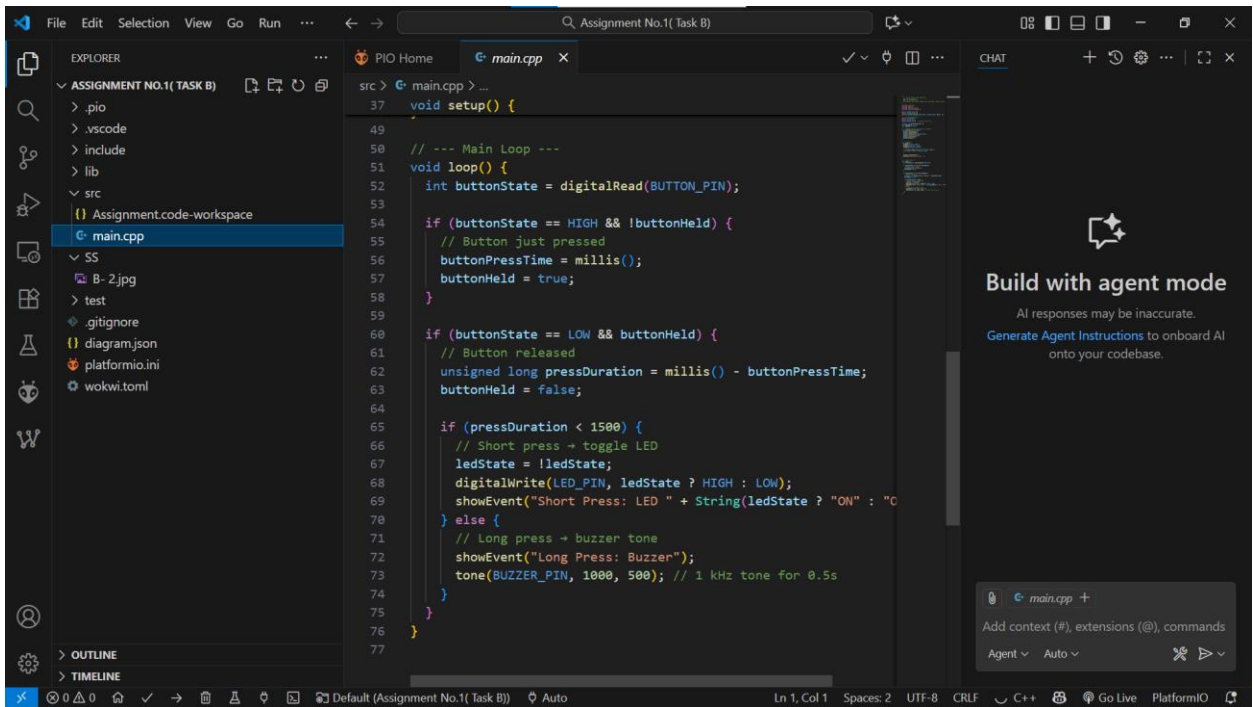
Generate Agent Instructions to onboard AI onto your codebase.

main.cpp +

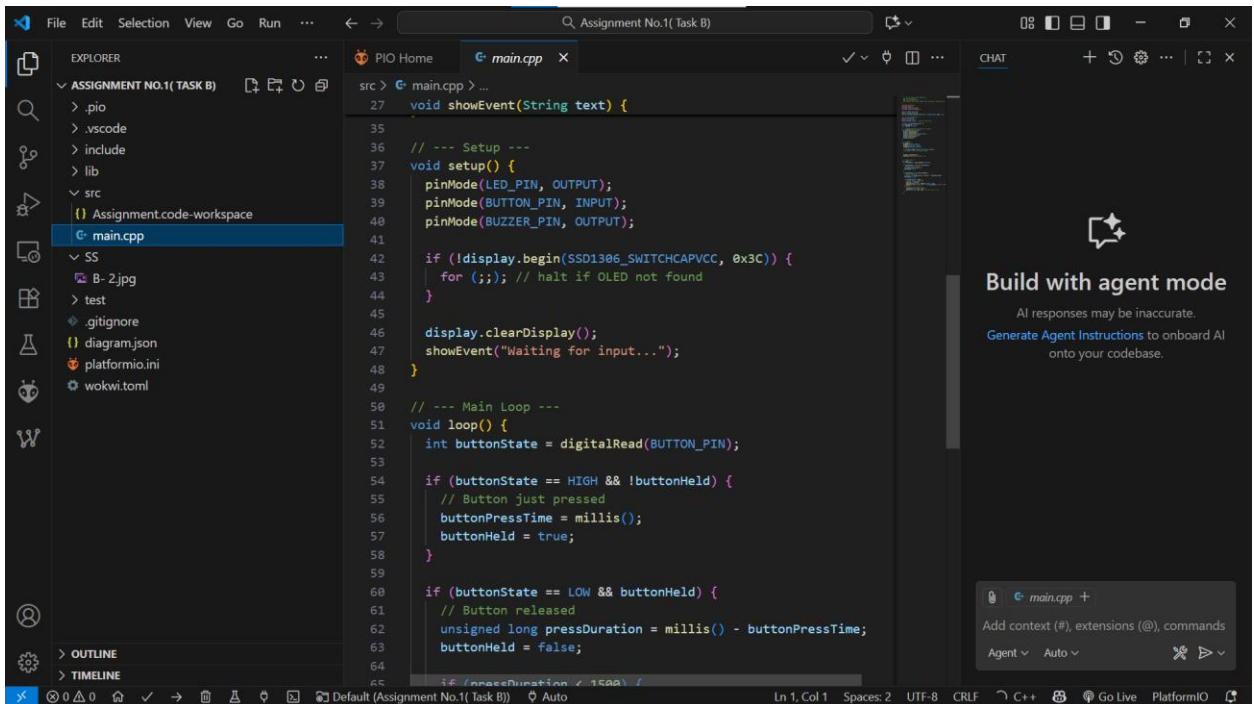
Add context (#), extensions (@), commands

Agent

Ln 1, Col 1 Spaces: 2 UTF-8 CRLF {} C++



```
src > main.cpp > ...
37 void setup() {
49
50 // --- Main Loop ---
51 void loop() {
52   int buttonState = digitalRead(BUTTON_PIN);
53
54   if (buttonState == HIGH && !buttonHeld) {
55     // Button just pressed
56     buttonPressTime = millis();
57     buttonHeld = true;
58   }
59
60   if (buttonState == LOW && buttonHeld) {
61     // Button released
62     unsigned long pressDuration = millis() - buttonPressTime;
63     buttonHeld = false;
64
65     if (pressDuration < 1500) {
66       // Short press -> toggle LED
67       ledState = !ledState;
68       digitalWrite(LED_PIN, ledState ? HIGH : LOW);
69       showEvent("Short Press: LED " + String(ledState ? "ON" : "C
70     } else {
71       // Long press -> buzzer tone
72       showEvent("Long Press: Buzzer");
73       tone(BUZZER_PIN, 1000, 500); // 1 kHz tone for 0.5s
74     }
75   }
76 }
77
```



```
src > main.cpp > ...
27 void showEvent(String text) {
36
37 // --- Setup ---
38 void setup() {
39   pinMode(LED_PIN, OUTPUT);
40   pinMode(BUTTON_PIN, INPUT);
41   pinMode(BUZZER_PIN, OUTPUT);
42
43   if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
44     for (;;) // halt if OLED not found
45   }
46
47   display.clearDisplay();
48   showEvent("Waiting for input...");
49
50 // --- Main Loop ---
51 void loop() {
52   int buttonState = digitalRead(BUTTON_PIN);
53
54   if (buttonState == HIGH && !buttonHeld) {
55     // Button just pressed
56     buttonPressTime = millis();
57     buttonHeld = true;
58   }
59
60   if (buttonState == LOW && buttonHeld) {
61     // Button released
62     unsigned long pressDuration = millis() - buttonPressTime;
63     buttonHeld = false;
64
65     if (pressDuration < 1500) {
66
```

➤ Pin Map:

- - LED1 → GPIO 18
- - Button → GPIO 35
- - Buzzer → GPIO 19
- - OLED → SDA 21, SCL 22

➤ Screenshots:

